

Ejercicios examen

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Ejercicios tipo examen

Estos son 4 ejercicios con diferentes bases de datos, lo primero que se hace es cargar todas las librerías necesarias y posteriormente se empezará con los ejercicios.

EJERCICIO 1

La primera serie de datos que vamos a emplear es USairpollution. Queremos saber cuáles son los mejores indicadores o predictores del grado de contaminación atmosférica en una ciudad, medido según el contenido de dióxido de azufre (SO₂) en el aire.

Primero cargamos la base de datos y la asignamos a un objeto, en este caso lo que pide el ejercicio es encontrar cuales son los mejores indicadores de dióxido de azufre atmosférico, para esto se realizará un análisis de regresión lineal múltiple, esto porque las instrucciones piden evaluar el comportamiento de una variable específica que en este caso es la concentración de *SO₂* la cual va a depender de todas las demás variables, además la regresión nos deja evaluar las contribuciones que tienen las demás variables sobre la variable de respuesta y se pueden descartar del modelo aquellas que no sean significativas o generen resultados extraños.

Primero se realizará un análisis exploratorio en donde se obtendrán las medidas centrales y de dispersión de la base de datos y después se hará el análisis ya descrito.

Resumen numérico y gráfico

```
summary(aire)
```

```
##          S02              temp              manu              popul
## Min.      : 8.00    Min.      :43.50    Min.      : 35.0    Min.      : 71.0
## 1st Qu.: 13.00    1st Qu.:50.60    1st Qu.: 181.0    1st Qu.: 299.0
## Median : 26.00    Median :54.60    Median : 347.0    Median : 515.0
## Mean      : 30.05    Mean      :55.76    Mean      : 463.1    Mean      : 608.6
## 3rd Qu.: 35.00    3rd Qu.:59.30    3rd Qu.: 462.0    3rd Qu.: 717.0
## Max.      :110.00    Max.      :75.50    Max.      :3344.0    Max.      :3369.0
##          wind          precip          predays
## Min.      : 6.000    Min.      : 7.05    Min.      : 36.0
## 1st Qu.: 8.700    1st Qu.:30.96    1st Qu.:103.0
## Median : 9.300    Median :38.74    Median :115.0
## Mean      : 9.444    Mean      :36.77    Mean      :113.9
## 3rd Qu.:10.600    3rd Qu.:43.11    3rd Qu.:128.0
## Max.      :12.700    Max.      :59.80    Max.      :166.0
```

```
tapply(aire,aire$manu,summary)
```

```
## $'35'
##      S02      temp      manu      popul      wind
## Min.   :31  Min.   :55.2  Min.   :35  Min.   :71  Min.   :6.5
## 1st Qu.:31  1st Qu.:55.2  1st Qu.:35  1st Qu.:71  1st Qu.:6.5
## Median :31  Median :55.2  Median :35  Median :71  Median :6.5
## Mean   :31  Mean   :55.2  Mean   :35  Mean   :71  Mean   :6.5
## 3rd Qu.:31  3rd Qu.:55.2  3rd Qu.:35  3rd Qu.:71  3rd Qu.:6.5
## Max.   :31  Max.   :55.2  Max.   :35  Max.   :71  Max.   :6.5
##      precip      predays
## Min.   :40.75  Min.   :148
## 1st Qu.:40.75  1st Qu.:148
## Median :40.75  Median :148
## Mean   :40.75  Mean   :148
## 3rd Qu.:40.75  3rd Qu.:148
## Max.   :40.75  Max.   :148
##
## $'44'
##      S02      temp      manu      popul      wind
## Min.   :46  Min.   :47.6  Min.   :44  Min.   :116  Min.   :8.8
## 1st Qu.:46  1st Qu.:47.6  1st Qu.:44  1st Qu.:116  1st Qu.:8.8
## Median :46  Median :47.6  Median :44  Median :116  Median :8.8
## Mean   :46  Mean   :47.6  Mean   :44  Mean   :116  Mean   :8.8
## 3rd Qu.:46  3rd Qu.:47.6  3rd Qu.:44  3rd Qu.:116  3rd Qu.:8.8
## Max.   :46  Max.   :47.6  Max.   :44  Max.   :116  Max.   :8.8
##      precip      predays
## Min.   :33.36  Min.   :135
## 1st Qu.:33.36  1st Qu.:135
## Median :33.36  Median :135
## Mean   :33.36  Mean   :135
## 3rd Qu.:33.36  3rd Qu.:135
## Max.   :33.36  Max.   :135
##
## $'46'
##      S02      temp      manu      popul      wind
## Min.   :11  Min.   :56.8  Min.   :46  Min.   :244  Min.   :8.9
## 1st Qu.:11  1st Qu.:56.8  1st Qu.:46  1st Qu.:244  1st Qu.:8.9
## Median :11  Median :56.8  Median :46  Median :244  Median :8.9
## Mean   :11  Mean   :56.8  Mean   :46  Mean   :244  Mean   :8.9
## 3rd Qu.:11  3rd Qu.:56.8  3rd Qu.:46  3rd Qu.:244  3rd Qu.:8.9
## Max.   :11  Max.   :56.8  Max.   :46  Max.   :244  Max.   :8.9
##      precip      predays
## Min.   :7.77  Min.   :58
## 1st Qu.:7.77  1st Qu.:58
## Median :7.77  Median :58
## Mean   :7.77  Mean   :58
## 3rd Qu.:7.77  3rd Qu.:58
## Max.   :7.77  Max.   :58
##
## $'80'
##      S02      temp      manu      popul      wind
## Min.   :36  Min.   :54  Min.   :80  Min.   :80  Min.   :9
```

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## 1st Qu.:36 1st Qu.:54 1st Qu.:80 1st Qu.:80 1st Qu.:9
## Median :36 Median :54 Median :80 Median :80 Median :9
## Mean :36 Mean :54 Mean :80 Mean :80 Mean :9
## 3rd Qu.:36 3rd Qu.:54 3rd Qu.:80 3rd Qu.:80 3rd Qu.:9
## Max. :36 Max. :54 Max. :80 Max. :80 Max. :9
## precip predays
## Min. :40.25 Min. :114
## 1st Qu.:40.25 1st Qu.:114
## Median :40.25 Median :114
## Mean :40.25 Mean :114
## 3rd Qu.:40.25 3rd Qu.:114
## Max. :40.25 Max. :114
##
## $'91'
## S02 temp manu popul wind
## Min. :13 Min. :61 Min. :91 Min. :132 Min. :8.2
## 1st Qu.:13 1st Qu.:61 1st Qu.:91 1st Qu.:132 1st Qu.:8.2
## Median :13 Median :61 Median :91 Median :132 Median :8.2
## Mean :13 Mean :61 Mean :91 Mean :132 Mean :8.2
## 3rd Qu.:13 3rd Qu.:61 3rd Qu.:91 3rd Qu.:132 3rd Qu.:8.2
## Max. :13 Max. :61 Max. :91 Max. :132 Max. :8.2
## precip predays
## Min. :48.52 Min. :100
## 1st Qu.:48.52 1st Qu.:100
## Median :48.52 Median :100
## Mean :48.52 Mean :100
## 3rd Qu.:48.52 3rd Qu.:100
## Max. :48.52 Max. :100
##
## $'96'
## S02 temp manu popul wind
## Min. :31 Min. :59.3 Min. :96 Min. :308 Min. :10.6
## 1st Qu.:31 1st Qu.:59.3 1st Qu.:96 1st Qu.:308 1st Qu.:10.6
## Median :31 Median :59.3 Median :96 Median :308 Median :10.6
## Mean :31 Mean :59.3 Mean :96 Mean :308 Mean :10.6
## 3rd Qu.:31 3rd Qu.:59.3 3rd Qu.:96 3rd Qu.:308 3rd Qu.:10.6
## Max. :31 Max. :59.3 Max. :96 Max. :308 Max. :10.6
## precip predays
## Min. :44.68 Min. :116
## 1st Qu.:44.68 1st Qu.:116
## Median :44.68 Median :116
## Mean :44.68 Mean :116
## 3rd Qu.:44.68 3rd Qu.:116
## Max. :44.68 Max. :116
##
## $'104'
## S02 temp manu popul wind
## Min. :17 Min. :49 Min. :104 Min. :201 Min. :11.2
## 1st Qu.:17 1st Qu.:49 1st Qu.:104 1st Qu.:201 1st Qu.:11.2
## Median :17 Median :49 Median :104 Median :201 Median :11.2
## Mean :17 Mean :49 Mean :104 Mean :201 Mean :11.2
## 3rd Qu.:17 3rd Qu.:49 3rd Qu.:104 3rd Qu.:201 3rd Qu.:11.2
## Max. :17 Max. :49 Max. :104 Max. :201 Max. :11.2
## precip predays

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## Min. :30.85 Min. :103
## 1st Qu.:30.85 1st Qu.:103
## Median :30.85 Median :103
## Mean :30.85 Mean :103
## 3rd Qu.:30.85 3rd Qu.:103
## Max. :30.85 Max. :103
##
## $'125'
## S02 temp manu popul wind
## Min. :8 Min. :56.6 Min. :125 Min. :277 Min. :12.7
## 1st Qu.:8 1st Qu.:56.6 1st Qu.:125 1st Qu.:277 1st Qu.:12.7
## Median :8 Median :56.6 Median :125 Median :277 Median :12.7
## Mean :8 Mean :56.6 Mean :125 Mean :277 Mean :12.7
## 3rd Qu.:8 3rd Qu.:56.6 3rd Qu.:125 3rd Qu.:277 3rd Qu.:12.7
## Max. :8 Max. :56.6 Max. :125 Max. :277 Max. :12.7
## precip predays
## Min. :30.58 Min. :82
## 1st Qu.:30.58 1st Qu.:82
## Median :30.58 Median :82
## Mean :30.58 Mean :82
## 3rd Qu.:30.58 3rd Qu.:82
## Max. :30.58 Max. :82
##
## $'136'
## S02 temp manu popul wind
## Min. :14 Min. :68.4 Min. :136 Min. :529 Min. :8.8
## 1st Qu.:14 1st Qu.:68.4 1st Qu.:136 1st Qu.:529 1st Qu.:8.8
## Median :14 Median :68.4 Median :136 Median :529 Median :8.8
## Mean :14 Mean :68.4 Mean :136 Mean :529 Mean :8.8
## 3rd Qu.:14 3rd Qu.:68.4 3rd Qu.:136 3rd Qu.:529 3rd Qu.:8.8
## Max. :14 Max. :68.4 Max. :136 Max. :529 Max. :8.8
## precip predays
## Min. :54.47 Min. :116
## 1st Qu.:54.47 1st Qu.:116
## Median :54.47 Median :116
## Mean :54.47 Mean :116
## 3rd Qu.:54.47 3rd Qu.:116
## Max. :54.47 Max. :116
##
## $'137'
## S02 temp manu popul wind
## Min. :28 Min. :51 Min. :137 Min. :176 Min. :8.7
## 1st Qu.:28 1st Qu.:51 1st Qu.:137 1st Qu.:176 1st Qu.:8.7
## Median :28 Median :51 Median :137 Median :176 Median :8.7
## Mean :28 Mean :51 Mean :137 Mean :176 Mean :8.7
## 3rd Qu.:28 3rd Qu.:51 3rd Qu.:137 3rd Qu.:176 3rd Qu.:8.7
## Max. :28 Max. :51 Max. :137 Max. :176 Max. :8.7
## precip predays
## Min. :15.17 Min. :89
## 1st Qu.:15.17 1st Qu.:89
## Median :15.17 Median :89
## Mean :15.17 Mean :89
## 3rd Qu.:15.17 3rd Qu.:89
## Max. :15.17 Max. :89

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##
## $'181'
##      S02      temp      manu      popul      wind
## Min.   :14   Min.   :51.5   Min.   :181   Min.   :347   Min.   :10.9
## 1st Qu.:14   1st Qu.:51.5   1st Qu.:181   1st Qu.:347   1st Qu.:10.9
## Median :14   Median :51.5   Median :181   Median :347   Median :10.9
## Mean   :14   Mean   :51.5   Mean   :181   Mean   :347   Mean   :10.9
## 3rd Qu.:14   3rd Qu.:51.5   3rd Qu.:181   3rd Qu.:347   3rd Qu.:10.9
## Max.   :14   Max.   :51.5   Max.   :181   Max.   :347   Max.   :10.9
##      precip      predays
## Min.   :30.18   Min.   :98
## 1st Qu.:30.18   1st Qu.:98
## Median :30.18   Median :98
## Mean   :30.18   Mean   :98
## 3rd Qu.:30.18   3rd Qu.:98
## Max.   :30.18   Max.   :98
##
## $'197'
##      S02      temp      manu      popul      wind
## Min.   :26   Min.   :57.8   Min.   :197   Min.   :299   Min.   :7.6
## 1st Qu.:26   1st Qu.:57.8   1st Qu.:197   1st Qu.:299   1st Qu.:7.6
## Median :26   Median :57.8   Median :197   Median :299   Median :7.6
## Mean   :26   Mean   :57.8   Mean   :197   Mean   :299   Mean   :7.6
## 3rd Qu.:26   3rd Qu.:57.8   3rd Qu.:197   3rd Qu.:299   3rd Qu.:7.6
## Max.   :26   Max.   :57.8   Max.   :197   Max.   :299   Max.   :7.6
##      precip      predays
## Min.   :42.59   Min.   :115
## 1st Qu.:42.59   1st Qu.:115
## Median :42.59   Median :115
## Mean   :42.59   Mean   :115
## 3rd Qu.:42.59   3rd Qu.:115
## Max.   :42.59   Max.   :115
##
## $'204'
##      S02      temp      manu      popul      wind
## Min.   :9   Min.   :68.3   Min.   :204   Min.   :361   Min.   :8.4
## 1st Qu.:9   1st Qu.:68.3   1st Qu.:204   1st Qu.:361   1st Qu.:8.4
## Median :9   Median :68.3   Median :204   Median :361   Median :8.4
## Mean   :9   Mean   :68.3   Mean   :204   Mean   :361   Mean   :8.4
## 3rd Qu.:9   3rd Qu.:68.3   3rd Qu.:204   3rd Qu.:361   3rd Qu.:8.4
## Max.   :9   Max.   :68.3   Max.   :204   Max.   :361   Max.   :8.4
##      precip      predays
## Min.   :56.77   Min.   :113
## 1st Qu.:56.77   1st Qu.:113
## Median :56.77   Median :113
## Mean   :56.77   Mean   :113
## 3rd Qu.:56.77   3rd Qu.:113
## Max.   :56.77   Max.   :113
##
## $'207'
##      S02      temp      manu      popul      wind
## Min.   :10   Min.   :75.5   Min.   :207   Min.   :335   Min.   :9
## 1st Qu.:10   1st Qu.:75.5   1st Qu.:207   1st Qu.:335   1st Qu.:9
## Median :10   Median :75.5   Median :207   Median :335   Median :9

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## Mean :10 Mean :75.5 Mean :207 Mean :335 Mean :9
## 3rd Qu.:10 3rd Qu.:75.5 3rd Qu.:207 3rd Qu.:335 3rd Qu.:9
## Max. :10 Max. :75.5 Max. :207 Max. :335 Max. :9
## precip predays
## Min. :59.8 Min. :128
## 1st Qu.:59.8 1st Qu.:128
## Median :59.8 Median :128
## Mean :59.8 Mean :128
## 3rd Qu.:59.8 3rd Qu.:128
## Max. :59.8 Max. :128
##
## $'213'
## S02 temp manu popul wind
## Min. :10 Min. :70.3 Min. :213 Min. :582 Min. :6
## 1st Qu.:10 1st Qu.:70.3 1st Qu.:213 1st Qu.:582 1st Qu.:6
## Median :10 Median :70.3 Median :213 Median :582 Median :6
## Mean :10 Mean :70.3 Mean :213 Mean :582 Mean :6
## 3rd Qu.:10 3rd Qu.:70.3 3rd Qu.:213 3rd Qu.:582 3rd Qu.:6
## Max. :10 Max. :70.3 Max. :213 Max. :582 Max. :6
## precip predays
## Min. :7.05 Min. :36
## 1st Qu.:7.05 1st Qu.:36
## Median :7.05 Median :36
## Mean :7.05 Mean :36
## 3rd Qu.:7.05 3rd Qu.:36
## Max. :7.05 Max. :36
##
## $'266'
## S02 temp manu popul wind
## Min. :26 Min. :51.5 Min. :266 Min. :540 Min. :8.6
## 1st Qu.:26 1st Qu.:51.5 1st Qu.:266 1st Qu.:540 1st Qu.:8.6
## Median :26 Median :51.5 Median :266 Median :540 Median :8.6
## Mean :26 Mean :51.5 Mean :266 Mean :540 Mean :8.6
## 3rd Qu.:26 3rd Qu.:51.5 3rd Qu.:266 3rd Qu.:540 3rd Qu.:8.6
## Max. :26 Max. :51.5 Max. :266 Max. :540 Max. :8.6
## precip predays
## Min. :37.01 Min. :134
## 1st Qu.:37.01 1st Qu.:134
## Median :37.01 Median :134
## Mean :37.01 Mean :134
## 3rd Qu.:37.01 3rd Qu.:134
## Max. :37.01 Max. :134
##
## $'275'
## S02 temp manu popul wind
## Min. :18 Min. :59.4 Min. :275 Min. :448 Min. :7.9
## 1st Qu.:18 1st Qu.:59.4 1st Qu.:275 1st Qu.:448 1st Qu.:7.9
## Median :18 Median :59.4 Median :275 Median :448 Median :7.9
## Mean :18 Mean :59.4 Mean :275 Mean :448 Mean :7.9
## 3rd Qu.:18 3rd Qu.:59.4 3rd Qu.:275 3rd Qu.:448 3rd Qu.:7.9
## Max. :18 Max. :59.4 Max. :275 Max. :448 Max. :7.9
## precip predays
## Min. :46 Min. :119
## 1st Qu.:46 1st Qu.:119

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## Median :46   Median :119
## Mean    :46   Mean    :119
## 3rd Qu. :46   3rd Qu. :119
## Max.    :46   Max.    :119
##
## $'291'
##      S02      temp      manu      popul      wind
## Min.   :30   Min.   :55.6   Min.   :291   Min.   :593   Min.   :8.3
## 1st Qu.:30   1st Qu.:55.6   1st Qu.:291   1st Qu.:593   1st Qu.:8.3
## Median :30   Median :55.6   Median :291   Median :593   Median :8.3
## Mean   :30   Mean   :55.6   Mean   :291   Mean   :593   Mean   :8.3
## 3rd Qu.:30   3rd Qu.:55.6   3rd Qu.:291   3rd Qu.:593   3rd Qu.:8.3
## Max.   :30   Max.   :55.6   Max.   :291   Max.   :593   Max.   :8.3
##      precip      predays
## Min.   :43.11   Min.   :123
## 1st Qu.:43.11   1st Qu.:123
## Median :43.11   Median :123
## Mean   :43.11   Mean   :123
## 3rd Qu.:43.11   3rd Qu.:123
## Max.   :43.11   Max.   :123
##
## $'337'
##      S02      temp      manu      popul      wind
## Min.   :10   Min.   :61.6   Min.   :337   Min.   :624   Min.   :9.2
## 1st Qu.:10   1st Qu.:61.6   1st Qu.:337   1st Qu.:624   1st Qu.:9.2
## Median :10   Median :61.6   Median :337   Median :624   Median :9.2
## Mean   :10   Mean   :61.6   Mean   :337   Mean   :624   Mean   :9.2
## 3rd Qu.:10   3rd Qu.:61.6   3rd Qu.:337   3rd Qu.:624   3rd Qu.:9.2
## Max.   :10   Max.   :61.6   Max.   :337   Max.   :624   Max.   :9.2
##      precip      predays
## Min.   :49.1   Min.   :105
## 1st Qu.:49.1   1st Qu.:105
## Median :49.1   Median :105
## Mean   :49.1   Mean   :105
## 3rd Qu.:49.1   3rd Qu.:105
## Max.   :49.1   Max.   :105
##
## $'343'
##      S02      temp      manu      popul      wind
## Min.   :94   Min.   :50   Min.   :343   Min.   :179   Min.   :10.6
## 1st Qu.:94   1st Qu.:50   1st Qu.:343   1st Qu.:179   1st Qu.:10.6
## Median :94   Median :50   Median :343   Median :179   Median :10.6
## Mean   :94   Mean   :50   Mean   :343   Mean   :179   Mean   :10.6
## 3rd Qu.:94   3rd Qu.:50   3rd Qu.:343   3rd Qu.:179   3rd Qu.:10.6
## Max.   :94   Max.   :50   Max.   :343   Max.   :179   Max.   :10.6
##      precip      predays
## Min.   :42.75   Min.   :125
## 1st Qu.:42.75   1st Qu.:125
## Median :42.75   Median :125
## Mean   :42.75   Mean   :125
## 3rd Qu.:42.75   3rd Qu.:125
## Max.   :42.75   Max.   :125
##
## $'347'

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```

##          S02          temp          manu          popul          wind
## Min.      :61   Min.      :50.4   Min.      :347   Min.      :520   Min.      :9.4
## 1st Qu.:61   1st Qu.:50.4   1st Qu.:347   1st Qu.:520   1st Qu.:9.4
## Median :61   Median :50.4   Median :347   Median :520   Median :9.4
## Mean      :61   Mean      :50.4   Mean      :347   Mean      :520   Mean      :9.4
## 3rd Qu.:61   3rd Qu.:50.4   3rd Qu.:347   3rd Qu.:520   3rd Qu.:9.4
## Max.      :61   Max.      :50.4   Max.      :347   Max.      :520   Max.      :9.4
##          precip          predays
## Min.      :36.22   Min.      :147
## 1st Qu.:36.22   1st Qu.:147
## Median :36.22   Median :147
## Mean      :36.22   Mean      :147
## 3rd Qu.:36.22   3rd Qu.:147
## Max.      :36.22   Max.      :147
##
## $'361'
##          S02          temp          manu          popul          wind
## Min.      :28   Min.      :52.3   Min.      :361   Min.      :746   Min.      :9.7
## 1st Qu.:28   1st Qu.:52.3   1st Qu.:361   1st Qu.:746   1st Qu.:9.7
## Median :28   Median :52.3   Median :361   Median :746   Median :9.7
## Mean      :28   Mean      :52.3   Mean      :361   Mean      :746   Mean      :9.7
## 3rd Qu.:28   3rd Qu.:52.3   3rd Qu.:361   3rd Qu.:746   3rd Qu.:9.7
## Max.      :28   Max.      :52.3   Max.      :361   Max.      :746   Max.      :9.7
##          precip          predays
## Min.      :38.74   Min.      :121
## 1st Qu.:38.74   1st Qu.:121
## Median :38.74   Median :121
## Mean      :38.74   Mean      :121
## 3rd Qu.:38.74   3rd Qu.:121
## Max.      :38.74   Max.      :121
##
## $'368'
##          S02          temp          manu          popul          wind
## Min.      :24   Min.      :61.5   Min.      :368   Min.      :497   Min.      :9.1
## 1st Qu.:24   1st Qu.:61.5   1st Qu.:368   1st Qu.:497   1st Qu.:9.1
## Median :24   Median :61.5   Median :368   Median :497   Median :9.1
## Mean      :24   Mean      :61.5   Mean      :368   Mean      :497   Mean      :9.1
## 3rd Qu.:24   3rd Qu.:61.5   3rd Qu.:368   3rd Qu.:497   3rd Qu.:9.1
## Max.      :24   Max.      :61.5   Max.      :368   Max.      :497   Max.      :9.1
##          precip          predays
## Min.      :48.34   Min.      :115
## 1st Qu.:48.34   1st Qu.:115
## Median :48.34   Median :115
## Mean      :48.34   Mean      :115
## 3rd Qu.:48.34   3rd Qu.:115
## Max.      :48.34   Max.      :115
##
## $'379'
##          S02          temp          manu          popul          wind
## Min.      :29   Min.      :51.1   Min.      :379   Min.      :531   Min.      :9.4
## 1st Qu.:29   1st Qu.:51.1   1st Qu.:379   1st Qu.:531   1st Qu.:9.4
## Median :29   Median :51.1   Median :379   Median :531   Median :9.4
## Mean      :29   Mean      :51.1   Mean      :379   Mean      :531   Mean      :9.4
## 3rd Qu.:29   3rd Qu.:51.1   3rd Qu.:379   3rd Qu.:531   3rd Qu.:9.4

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## Max. :29 Max. :51.1 Max. :379 Max. :531 Max. :9.4
## precip predays
## Min. :38.79 Min. :164
## 1st Qu.:38.79 1st Qu.:164
## Median :38.79 Median :164
## Mean :38.79 Mean :164
## 3rd Qu.:38.79 3rd Qu.:164
## Max. :38.79 Max. :164
##
## $'381'
## S02 temp manu popul wind
## Min. :14 Min. :54.5 Min. :381 Min. :507 Min. :10
## 1st Qu.:14 1st Qu.:54.5 1st Qu.:381 1st Qu.:507 1st Qu.:10
## Median :14 Median :54.5 Median :381 Median :507 Median :10
## Mean :14 Mean :54.5 Mean :381 Mean :507 Mean :10
## 3rd Qu.:14 3rd Qu.:54.5 3rd Qu.:381 3rd Qu.:507 3rd Qu.:10
## Max. :14 Max. :54.5 Max. :381 Max. :507 Max. :10
## precip predays
## Min. :37 Min. :99
## 1st Qu.:37 1st Qu.:99
## Median :37 Median :99
## Mean :37 Mean :99
## 3rd Qu.:37 3rd Qu.:99
## Max. :37 Max. :99
##
## $'391'
## S02 temp manu popul wind
## Min. :11 Min. :47.1 Min. :391 Min. :463 Min. :12.4
## 1st Qu.:11 1st Qu.:47.1 1st Qu.:391 1st Qu.:463 1st Qu.:12.4
## Median :11 Median :47.1 Median :391 Median :463 Median :12.4
## Mean :11 Mean :47.1 Mean :391 Mean :463 Mean :12.4
## 3rd Qu.:11 3rd Qu.:47.1 3rd Qu.:391 3rd Qu.:463 3rd Qu.:12.4
## Max. :11 Max. :47.1 Max. :391 Max. :463 Max. :12.4
## precip predays
## Min. :36.11 Min. :166
## 1st Qu.:36.11 1st Qu.:166
## Median :36.11 Median :166
## Mean :36.11 Mean :166
## 3rd Qu.:36.11 3rd Qu.:166
## Max. :36.11 Max. :166
##
## $'412'
## S02 temp manu popul wind
## Min. :56 Min. :49.1 Min. :412 Min. :158 Min. :9
## 1st Qu.:56 1st Qu.:49.1 1st Qu.:412 1st Qu.:158 1st Qu.:9
## Median :56 Median :49.1 Median :412 Median :158 Median :9
## Mean :56 Mean :49.1 Mean :412 Mean :158 Mean :9
## 3rd Qu.:56 3rd Qu.:49.1 3rd Qu.:412 3rd Qu.:158 3rd Qu.:9
## Max. :56 Max. :49.1 Max. :412 Max. :158 Max. :9
## precip predays
## Min. :43.37 Min. :127
## 1st Qu.:43.37 1st Qu.:127
## Median :43.37 Median :127
## Mean :43.37 Mean :127

```

```

## 3rd Qu.:43.37 3rd Qu.:127
## Max. :43.37 Max. :127
##
## $'434'
##      S02      temp      manu      popul      wind
## Min. :29 Min. :57.3 Min. :434 Min. :757 Min. :9.3
## 1st Qu.:29 1st Qu.:57.3 1st Qu.:434 1st Qu.:757 1st Qu.:9.3
## Median :29 Median :57.3 Median :434 Median :757 Median :9.3
## Mean :29 Mean :57.3 Mean :434 Mean :757 Mean :9.3
## 3rd Qu.:29 3rd Qu.:57.3 3rd Qu.:434 3rd Qu.:757 3rd Qu.:9.3
## Max. :29 Max. :57.3 Max. :434 Max. :757 Max. :9.3
##      precip      predays
## Min. :38.89 Min. :111
## 1st Qu.:38.89 1st Qu.:111
## Median :38.89 Median :111
## Mean :38.89 Mean :111
## 3rd Qu.:38.89 3rd Qu.:111
## Max. :38.89 Max. :111
##
## $'453'
##      S02      temp      manu      popul      wind
## Min. :12 Min. :56.7 Min. :453 Min. :716 Min. :8.7
## 1st Qu.:12 1st Qu.:56.7 1st Qu.:453 1st Qu.:716 1st Qu.:8.7
## Median :12 Median :56.7 Median :453 Median :716 Median :8.7
## Mean :12 Mean :56.7 Mean :453 Mean :716 Mean :8.7
## 3rd Qu.:12 3rd Qu.:56.7 3rd Qu.:453 3rd Qu.:716 3rd Qu.:8.7
## Max. :12 Max. :56.7 Max. :453 Max. :716 Max. :8.7
##      precip      predays
## Min. :20.66 Min. :67
## 1st Qu.:20.66 1st Qu.:67
## Median :20.66 Median :67
## Mean :20.66 Mean :67
## 3rd Qu.:20.66 3rd Qu.:67
## Max. :20.66 Max. :67
##
## $'454'
##      S02      temp      manu      popul      wind
## Min. :17 Min. :51.9 Min. :454 Min. :515 Min. :9
## 1st Qu.:17 1st Qu.:51.9 1st Qu.:454 1st Qu.:515 1st Qu.:9
## Median :17 Median :51.9 Median :454 Median :515 Median :9
## Mean :17 Mean :51.9 Mean :454 Mean :515 Mean :9
## 3rd Qu.:17 3rd Qu.:51.9 3rd Qu.:454 3rd Qu.:515 3rd Qu.:9
## Max. :17 Max. :51.9 Max. :454 Max. :515 Max. :9
##      precip      predays
## Min. :12.95 Min. :86
## 1st Qu.:12.95 1st Qu.:86
## Median :12.95 Median :86
## Mean :12.95 Mean :86
## 3rd Qu.:12.95 3rd Qu.:86
## Max. :12.95 Max. :86
##
## $'462'
##      S02      temp      manu      popul      wind
## Min. :23 Min. :54 Min. :462 Min. :453 Min. :7.1

```

```

## 1st Qu.:23 1st Qu.:54 1st Qu.:462 1st Qu.:453 1st Qu.:7.1
## Median :23 Median :54 Median :462 Median :453 Median :7.1
## Mean :23 Mean :54 Mean :462 Mean :453 Mean :7.1
## 3rd Qu.:23 3rd Qu.:54 3rd Qu.:462 3rd Qu.:453 3rd Qu.:7.1
## Max. :23 Max. :54 Max. :462 Max. :453 Max. :7.1
## precip predays
## Min. :39.04 Min. :132
## 1st Qu.:39.04 1st Qu.:132
## Median :39.04 Median :132
## Mean :39.04 Mean :132
## 3rd Qu.:39.04 3rd Qu.:132
## Max. :39.04 Max. :132
##
## $'569'
## S02 temp manu popul wind
## Min. :16 Min. :45.7 Min. :569 Min. :717 Min. :11.8
## 1st Qu.:16 1st Qu.:45.7 1st Qu.:569 1st Qu.:717 1st Qu.:11.8
## Median :16 Median :45.7 Median :569 Median :717 Median :11.8
## Mean :16 Mean :45.7 Mean :569 Mean :717 Mean :11.8
## 3rd Qu.:16 3rd Qu.:45.7 3rd Qu.:569 3rd Qu.:717 3rd Qu.:11.8
## Max. :16 Max. :45.7 Max. :569 Max. :717 Max. :11.8
## precip predays
## Min. :29.07 Min. :123
## 1st Qu.:29.07 1st Qu.:123
## Median :29.07 Median :123
## Mean :29.07 Mean :123
## 3rd Qu.:29.07 3rd Qu.:123
## Max. :29.07 Max. :123
##
## $'625'
## S02 temp manu popul wind
## Min. :47 Min. :55 Min. :625 Min. :905 Min. :9.6
## 1st Qu.:47 1st Qu.:55 1st Qu.:625 1st Qu.:905 1st Qu.:9.6
## Median :47 Median :55 Median :625 Median :905 Median :9.6
## Mean :47 Mean :55 Mean :625 Mean :905 Mean :9.6
## 3rd Qu.:47 3rd Qu.:55 3rd Qu.:625 3rd Qu.:905 3rd Qu.:9.6
## Max. :47 Max. :55 Max. :625 Max. :905 Max. :9.6
## precip predays
## Min. :41.31 Min. :111
## 1st Qu.:41.31 1st Qu.:111
## Median :41.31 Median :111
## Mean :41.31 Mean :111
## 3rd Qu.:41.31 3rd Qu.:111
## Max. :41.31 Max. :111
##
## $'641'
## S02 temp manu popul wind
## Min. :9 Min. :66.2 Min. :641 Min. :844 Min. :10.9
## 1st Qu.:9 1st Qu.:66.2 1st Qu.:641 1st Qu.:844 1st Qu.:10.9
## Median :9 Median :66.2 Median :641 Median :844 Median :10.9
## Mean :9 Mean :66.2 Mean :641 Mean :844 Mean :10.9
## 3rd Qu.:9 3rd Qu.:66.2 3rd Qu.:641 3rd Qu.:844 3rd Qu.:10.9
## Max. :9 Max. :66.2 Max. :641 Max. :844 Max. :10.9
## precip predays

```

```

## Min. :35.94 Min. :78
## 1st Qu.:35.94 1st Qu.:78
## Median :35.94 Median :78
## Mean :35.94 Mean :78
## 3rd Qu.:35.94 3rd Qu.:78
## Max. :35.94 Max. :78
##
## $'699'
## S02 temp manu popul wind
## Min. :29 Min. :43.5 Min. :699 Min. :744 Min. :10.6
## 1st Qu.:29 1st Qu.:43.5 1st Qu.:699 1st Qu.:744 1st Qu.:10.6
## Median :29 Median :43.5 Median :699 Median :744 Median :10.6
## Mean :29 Mean :43.5 Mean :699 Mean :744 Mean :10.6
## 3rd Qu.:29 3rd Qu.:43.5 3rd Qu.:699 3rd Qu.:744 3rd Qu.:10.6
## Max. :29 Max. :43.5 Max. :699 Max. :744 Max. :10.6
## precip predays
## Min. :25.94 Min. :137
## 1st Qu.:25.94 1st Qu.:137
## Median :25.94 Median :137
## Mean :25.94 Mean :137
## 3rd Qu.:25.94 3rd Qu.:137
## Max. :25.94 Max. :137
##
## $'721'
## S02 temp manu popul wind
## Min. :10 Min. :68.9 Min. :721 Min. :1233 Min. :10.8
## 1st Qu.:10 1st Qu.:68.9 1st Qu.:721 1st Qu.:1233 1st Qu.:10.8
## Median :10 Median :68.9 Median :721 Median :1233 Median :10.8
## Mean :10 Mean :68.9 Mean :721 Mean :1233 Mean :10.8
## 3rd Qu.:10 3rd Qu.:68.9 3rd Qu.:721 3rd Qu.:1233 3rd Qu.:10.8
## Max. :10 Max. :68.9 Max. :721 Max. :1233 Max. :10.8
## precip predays
## Min. :48.19 Min. :103
## 1st Qu.:48.19 1st Qu.:103
## Median :48.19 Median :103
## Mean :48.19 Mean :103
## 3rd Qu.:48.19 3rd Qu.:103
## Max. :48.19 Max. :103
##
## $'775'
## S02 temp manu popul wind
## Min. :56 Min. :55.9 Min. :775 Min. :622 Min. :9.5
## 1st Qu.:56 1st Qu.:55.9 1st Qu.:775 1st Qu.:622 1st Qu.:9.5
## Median :56 Median :55.9 Median :775 Median :622 Median :9.5
## Mean :56 Mean :55.9 Mean :775 Mean :622 Mean :9.5
## 3rd Qu.:56 3rd Qu.:55.9 3rd Qu.:775 3rd Qu.:622 3rd Qu.:9.5
## Max. :56 Max. :55.9 Max. :775 Max. :622 Max. :9.5
## precip predays
## Min. :35.89 Min. :105
## 1st Qu.:35.89 1st Qu.:105
## Median :35.89 Median :105
## Mean :35.89 Mean :105
## 3rd Qu.:35.89 3rd Qu.:105
## Max. :35.89 Max. :105

```

```

##
## $'1007'
##      S02      temp      manu      popul      wind
## Min.   :65   Min.   :49.7   Min.   :1007   Min.   :751   Min.   :10.9
## 1st Qu.:65   1st Qu.:49.7   1st Qu.:1007   1st Qu.:751   1st Qu.:10.9
## Median :65   Median :49.7   Median :1007   Median :751   Median :10.9
## Mean   :65   Mean   :49.7   Mean   :1007   Mean   :751   Mean   :10.9
## 3rd Qu.:65   3rd Qu.:49.7   3rd Qu.:1007   3rd Qu.:751   3rd Qu.:10.9
## Max.   :65   Max.   :49.7   Max.   :1007   Max.   :751   Max.   :10.9
##      precip      predays
## Min.   :34.99   Min.   :155
## 1st Qu.:34.99   1st Qu.:155
## Median :34.99   Median :155
## Mean   :34.99   Mean   :155
## 3rd Qu.:34.99   3rd Qu.:155
## Max.   :34.99   Max.   :155
##
## $'1064'
##      S02      temp      manu      popul      wind
## Min.   :35   Min.   :49.9   Min.   :1064   Min.   :1513   Min.   :10.1
## 1st Qu.:35   1st Qu.:49.9   1st Qu.:1064   1st Qu.:1513   1st Qu.:10.1
## Median :35   Median :49.9   Median :1064   Median :1513   Median :10.1
## Mean   :35   Mean   :49.9   Mean   :1064   Mean   :1513   Mean   :10.1
## 3rd Qu.:35   3rd Qu.:49.9   3rd Qu.:1064   3rd Qu.:1513   3rd Qu.:10.1
## Max.   :35   Max.   :49.9   Max.   :1064   Max.   :1513   Max.   :10.1
##      precip      predays
## Min.   :30.96   Min.   :129
## 1st Qu.:30.96   1st Qu.:129
## Median :30.96   Median :129
## Mean   :30.96   Mean   :129
## 3rd Qu.:30.96   3rd Qu.:129
## Max.   :30.96   Max.   :129
##
## $'1692'
##      S02      temp      manu      popul      wind
## Min.   :69   Min.   :54.6   Min.   :1692   Min.   :1950   Min.   :9.6
## 1st Qu.:69   1st Qu.:54.6   1st Qu.:1692   1st Qu.:1950   1st Qu.:9.6
## Median :69   Median :54.6   Median :1692   Median :1950   Median :9.6
## Mean   :69   Mean   :54.6   Mean   :1692   Mean   :1950   Mean   :9.6
## 3rd Qu.:69   3rd Qu.:54.6   3rd Qu.:1692   3rd Qu.:1950   3rd Qu.:9.6
## Max.   :69   Max.   :54.6   Max.   :1692   Max.   :1950   Max.   :9.6
##      precip      predays
## Min.   :39.93   Min.   :115
## 1st Qu.:39.93   1st Qu.:115
## Median :39.93   Median :115
## Mean   :39.93   Mean   :115
## 3rd Qu.:39.93   3rd Qu.:115
## Max.   :39.93   Max.   :115
##
## $'3344'
##      S02      temp      manu      popul      wind
## Min.   :110   Min.   :50.6   Min.   :3344   Min.   :3369   Min.   :10.4
## 1st Qu.:110   1st Qu.:50.6   1st Qu.:3344   1st Qu.:3369   1st Qu.:10.4
## Median :110   Median :50.6   Median :3344   Median :3369   Median :10.4

```

```
## Mean :110 Mean :50.6 Mean :3344 Mean :3369 Mean :10.4
## 3rd Qu.:110 3rd Qu.:50.6 3rd Qu.:3344 3rd Qu.:3369 3rd Qu.:10.4
## Max. :110 Max. :50.6 Max. :3344 Max. :3369 Max. :10.4
## precip predays
## Min. :34.44 Min. :122
## 1st Qu.:34.44 1st Qu.:122
## Median :34.44 Median :122
## Mean :34.44 Mean :122
## 3rd Qu.:34.44 3rd Qu.:122
## Max. :34.44 Max. :122
```

```
summarySE(aire,measurevar="S02")
```

```
## .id N S02 sd se ci
## 1 <NA> 41 30.04878 23.47227 3.665753 7.408763
```

```
summarySE(aire,measurevar="temp")
```

```
## .id N temp sd se ci
## 1 <NA> 41 55.76341 7.227716 1.12878 2.281349
```

```
summarySE(aire,measurevar="manu")
```

```
## .id N manu sd se ci
## 1 <NA> 41 463.0976 563.4739 87.99985 177.8543
```

```
summarySE(aire,measurevar="popul")
```

```
## .id N popul sd se ci
## 1 <NA> 41 608.6098 579.113 90.44226 182.7906
```

```
summarySE(aire,measurevar="wind")
```

```
## .id N wind sd se ci
## 1 <NA> 41 9.443902 1.428644 0.2231167 0.4509358
```

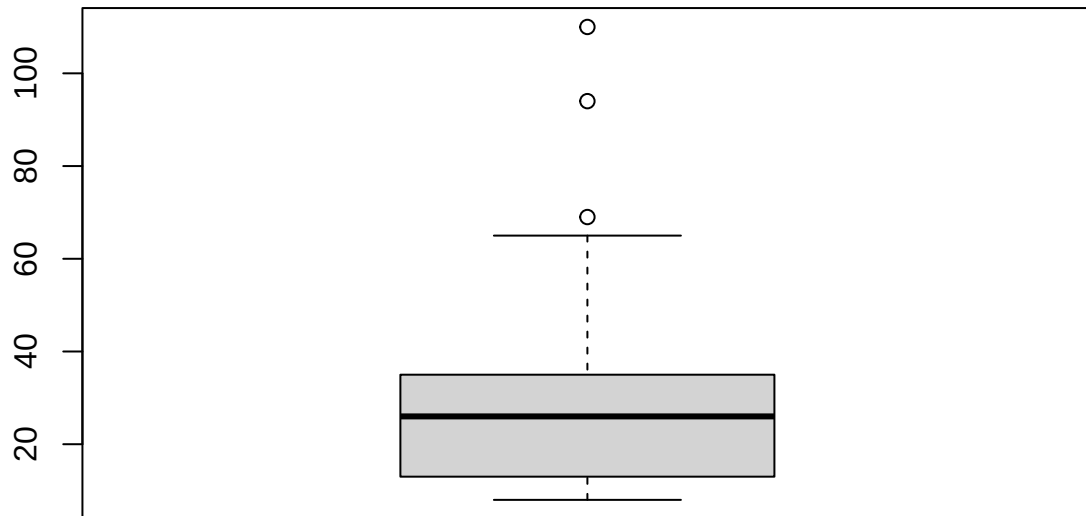
```
summarySE(aire,measurevar="precip")
```

```
## .id N precip sd se ci
## 1 <NA> 41 36.76902 11.77155 1.838407 3.71556
```

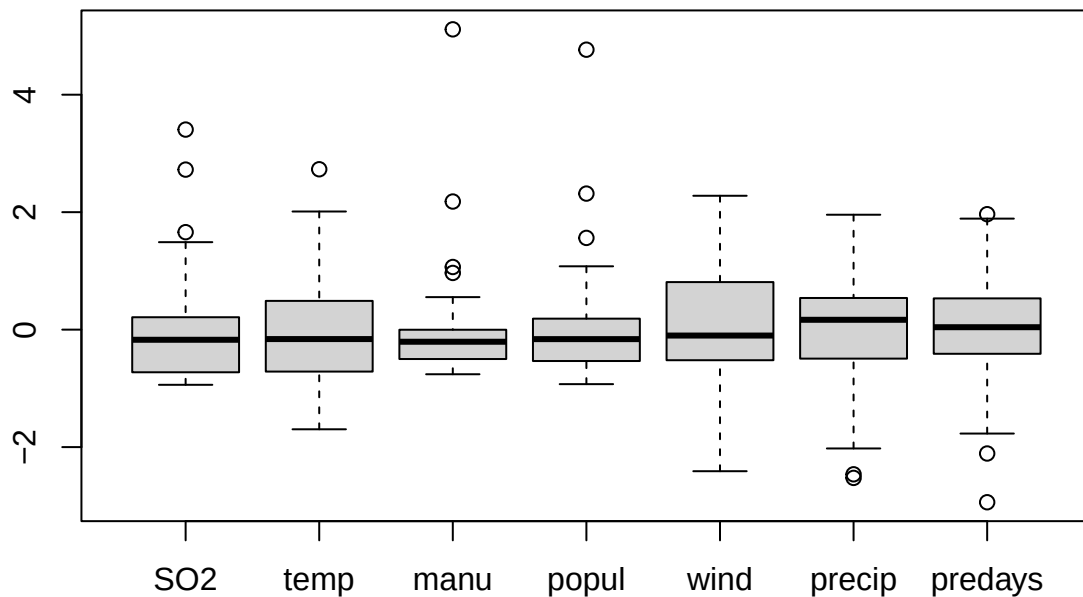
```
summarySE(aire,measurevar="predays")
```

```
## .id N predays sd se ci
## 1 <NA> 41 113.9024 26.50642 4.139607 8.366458
```

```
boxplot(aire$S02)
```



```
boxplot(scale(aire))
```



Se realizaron estas líneas de código con las funciones *summary* y *summarySE* para obtener las medidas centrales y de dispersión, la primera nos da un resumen de toda la base, pero es demasiado extenso, para eso se corrió la segunda línea de código la cual ejecuta la misma función pero agrupando los resultados bajo la variable de manufactura, es decir, la cantidad de fábricas. Después de esto con la función *summarySE* se llevó a cabo las medidas de dispersión de todas las variables.

Finalmente se realizó un boxplot único de la variable SO2 y otro de toda la base de datos escalada porque los datos de población eran demasiado grandes en comparación con las demás variables lo que hacía que las demás cajas se vieran aplastadas.

Regresión múltiple

```
ejercicio1<-lm(SO2~temp+manu+popul+wind+precip+predays,data=aire)
summary(ejercicio1)
```

```
##
## Call:
## lm(formula = SO2 ~ temp + manu + popul + wind + precip + predays,
##     data = aire)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.004  -8.542  -0.991   5.758  48.758
##
## Coefficients:
```



```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 111.72848   47.31810   2.361 0.024087 *
## temp        -1.26794    0.62118  -2.041 0.049056 *
## manu         0.06492    0.01575   4.122 0.000228 ***
## popul       -0.03928    0.01513  -2.595 0.013846 *
## wind        -3.18137    1.81502  -1.753 0.088650 .
## precip       0.51236    0.36276   1.412 0.166918
## predays     -0.05205    0.16201  -0.321 0.749972
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.64 on 34 degrees of freedom
## Multiple R-squared:  0.6695, Adjusted R-squared:  0.6112
## F-statistic: 11.48 on 6 and 34 DF,  p-value: 5.419e-07
```

```
ejercicio1.1<-step(ejercicio1)
```

```
## Start:  AIC=226.37
## S02 ~ temp + manu + popul + wind + precip + predays
##
##           Df Sum of Sq    RSS    AIC
## - predays  1      22.1  7305.4 224.50
## <none>                        7283.3 226.37
## - precip   1     427.3  7710.6 226.71
## - wind     1     658.1  7941.4 227.92
## - temp     1     892.5  8175.8 229.11
## - popul    1    1443.1  8726.3 231.78
## - manu     1    3640.1 10923.4 240.99
##
## Step:  AIC=224.49
## S02 ~ temp + manu + popul + wind + precip
##
##           Df Sum of Sq    RSS    AIC
## <none>                        7305.4 224.50
## - wind     1     636.1  7941.5 225.92
## - precip   1     785.4  8090.8 226.68
## - popul    1    1447.5  8752.9 229.91
## - temp     1    1517.4  8822.8 230.23
## - manu     1    3636.8 10942.1 239.06
```

```
summary(ejercicio1.1)
```

```
##
## Call:
## lm(formula = S02 ~ temp + manu + popul + wind + precip, data = aire)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.253  -7.655  -0.581   6.059  49.438
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 100.15245   30.27521   3.308 0.002182 **
```

```
## temp      -1.12129    0.41586   -2.696 0.010707 *
## manu       0.06489    0.01554    4.174 0.000188 ***
## popul     -0.03933    0.01494   -2.633 0.012499 *
## wind      -3.08240    1.76562   -1.746 0.089622 .
## precip     0.41947    0.21624    1.940 0.060498 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.45 on 35 degrees of freedom
## Multiple R-squared:  0.6685, Adjusted R-squared:  0.6212
## F-statistic: 14.12 on 5 and 35 DF,  p-value: 1.409e-07
```

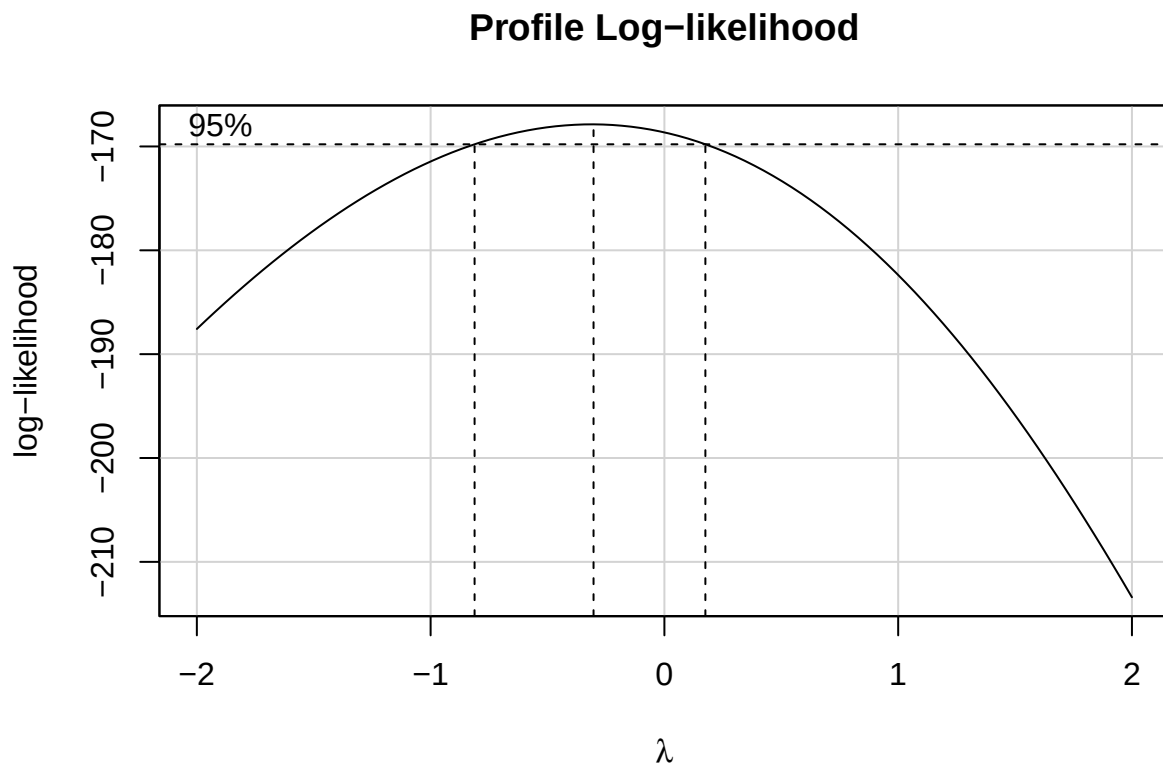
```
shapiro.test(ejercicio1.1$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  ejercicio1.1$residuals
## W = 0.92466, p-value = 0.009644
```

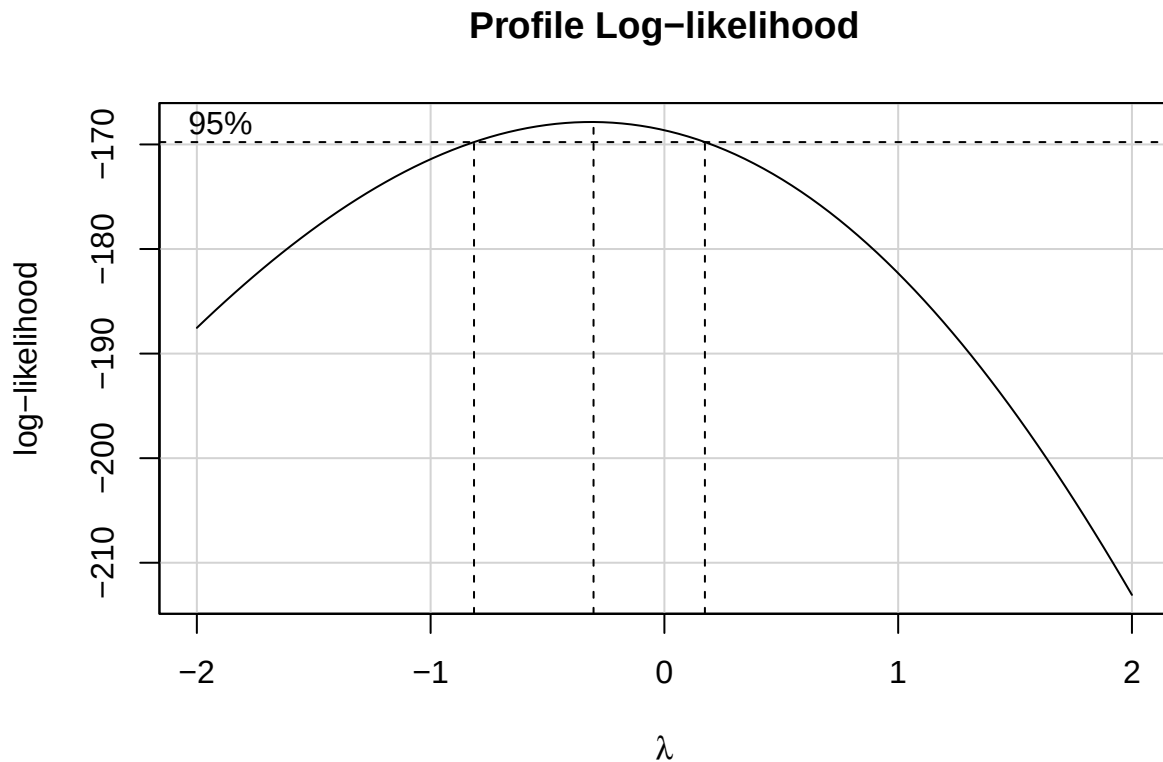
```
vif(ejercicio1.1)
```

```
##      temp      manu      popul      wind      precip
## 1.731366 14.703099 14.338797  1.219354  1.241777
```

```
boxCox(ejercicio1.1)
```



```
bc<-boxCox(ejercicio1)
```



```
lambda_optimo <- bc$x[which.max(bc$y)]
print(lambda_optimo)
```

```
## [1] -0.3030303
```

Se realizó el primer modelo, el cual incluye todas las variables, el resumen de este modelo nos indica que la variable *predays* no es significativa, por lo tanto se optó por usar la función *step* la cual eliminó dicha variable y en el resumen de este nuevo modelo se puede observar que el ajuste de la R cuadrada ajustada mejora por 1% y de igual manera el valor de akaike disminuye por lo tanto se puede concluir que esa variable no es significativa para el modelo.

Al revisar los supuestos del modelo se observa que no hay normalidad en los residuales del mismo, después de esto se usó la función *vif* para verificar la correlación entre variables lo cual permitió observar que las variables de manufactura y población están correlacionadas entre si.

Para tratar de corregir estas situaciones en los datos, se realiza la transformación de boxCox para ilustrar que tipo de transformación usar.

```
ejercicio1.2<-lm(log(SO2)~temp+manu+wind+precip,data=aire)
summary(ejercicio1.2)
```

```
##
## Call:
## lm(formula = log(SO2) ~ temp + manu + wind + precip, data = aire)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.77450 -0.26074 -0.05387  0.27927  1.14267
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.7628404  0.9032385   8.594 3.02e-10 ***
## temp        -0.0699392  0.0115861  -6.036 6.21e-07 ***
## manu         0.0005556  0.0001310   4.242 0.000148 ***
## wind        -0.1803936  0.0546244  -3.302 0.002172 **
## precip       0.0200318  0.0066221   3.025 0.004568 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4501 on 36 degrees of freedom
## Multiple R-squared:  0.6303, Adjusted R-squared:  0.5892
## F-statistic: 15.35 on 4 and 36 DF,  p-value: 2.053e-07
```

```
vif(ejercicio1.2)
```

```
##      temp      manu      wind      precip
## 1.384555 1.075237 1.202426 1.199748
```

```
shapiro.test(ejercicio1.2$residuals)
```

```
##
## Shapiro-Wilk normality test
##
## data:  ejercicio1.2$residuals
## W = 0.97936, p-value = 0.6515
```

```
ncvTest(ejercicio1.2)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 0.4177226, Df = 1, p = 0.51808
```

```
durbinWatsonTest(ejercicio1.2)
```

```
## lag Autocorrelation D-W Statistic p-value
## 1      0.03122061      1.907262      0.788
## Alternative hypothesis: rho != 0
```

```
ejercicio1.3<-lm(SO2~-0.3~temp+manu+wind+precip,data=aire)
summary(ejercicio1.3)
```

```
##
## Call:
## lm(formula = SO2~-0.3 ~ temp + manu + wind + precip, data = aire)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.110113 -0.030497  0.003142  0.028155  0.091679
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.859e-01  1.019e-01  -1.825  0.076307 .
## temp         8.609e-03  1.307e-03   6.587  1.15e-07 ***
## manu        -5.654e-05  1.477e-05  -3.827  0.000498 ***
## wind         2.290e-02  6.162e-03   3.717  0.000682 ***
## precip      -2.383e-03  7.470e-04  -3.191  0.002939 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05077 on 36 degrees of freedom
## Multiple R-squared:  0.6439, Adjusted R-squared:  0.6043
## F-statistic: 16.27 on 4 and 36 DF,  p-value: 1.068e-07
```

```
vif(ejercicio1.3)
```

```
##      temp      manu      wind      precip
## 1.384555 1.075237 1.202426 1.199748
```

```
shapiro.test(ejercicio1.3$residuals)
```

```
##
## Shapiro-Wilk normality test
##
## data:  ejercicio1.3$residuals
## W = 0.98645, p-value = 0.8989
```

```
ncvTest(ejercicio1.3)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 0.1066493, Df = 1, p = 0.74399
```

```
durbinWatsonTest(ejercicio1.3)
```

```
## lag Autocorrelation D-W Statistic p-value
## 1 -0.0171636 2.006615 0.94
## Alternative hypothesis: rho != 0
```

En base a estos resultados se hicieron 2 modelos más, uno logarítmico y otro con el valor que dio la transformación de boxCox del primer modelo, en ambos casos los resultados fueron muy similares y todos los supuestos se cumplieron obteniendo un ajuste del 58% y del 60% respectivamente.

Para este ejercicio, los resultados indican que la temperatura y el viento tienen un efecto de mitigación, por lo que a mayor temperatura o más velocidad del viento la concentración de SO₂ disminuye por 6% y 18% respectivamente, por otro lado la cantidad de fábricas y la precipitación parecen ser indicadores positivos, es decir a mayor precipitación y más cantidad de fábricas en la ciudad, la presencia del SO₂ en el aire aumenta 2% y 0.05% respectivamente.

Ejercicio 2

Primero cargar los datos y asignarlos

Para este ejercicio se hará un análisis de conglomerados, esto principalmente por la indicación que da el ejercicio el cual se basa en agrupar a los individuos de los datos y agruparlos en base al tipo de mujer. Lo primero que se hace es un análisis numérico y gráfico.

```
summary(ejercicio2)
```

```
##           R_region      R_age      R_edu
## Coastal South:319   Min.    :20.00   Never attended school: 90
## Coastal Easth:331  1st Qu.:32.00   Elementary school    :267
## Inlands           :156   Median  :38.00   Junior high school   :583
## North             :241   Mean     :38.99   Senior high school   :425
## Northeast         :279  3rd Qu.:45.00   Junior college       :125
## Central West :208   Max.     :64.00   University           : 44
##
##      R_income      R_health      R_height      R_happy
## Min.    :    0   Poor      : 10   Min.    :140.0   Very unhappy : 14
## 1st Qu.: 200   Not good :139   1st Qu.:156.0   Not too happy : 185
## Median : 500   Fair      :461   Median :160.0   Somewhat happy:1055
## Mean   : 617   Good      :582   Mean   :159.3   Very happy    : 280
## 3rd Qu.: 800   Excellent:342   3rd Qu.:162.8
## Max.   :10000                Max.   :178.0
##
##      A_height      A_edu      A_income
## Min.    :155.0   Never attended school: 30   Min.    :    0.0
## 1st Qu.:168.0   Elementary school    :204   1st Qu.: 400.0
## Median :170.0   Junior high school   :587   Median : 700.0
## Mean   :171.2   Senior high school   :464   Mean   : 986.7
## 3rd Qu.:175.0   Junior college       :146   3rd Qu.:1000.0
## Max.   :190.0   University           :100   Max.   :10000.0
##
##           NAs           : 3
```

```
summarySE(ejercicio2,measurevar = "R_age")
```

```
##      .id      N      R_age      sd      se      ci
## 1 <NA> 1534 38.99413 9.571748 0.2443873 0.4793687
```

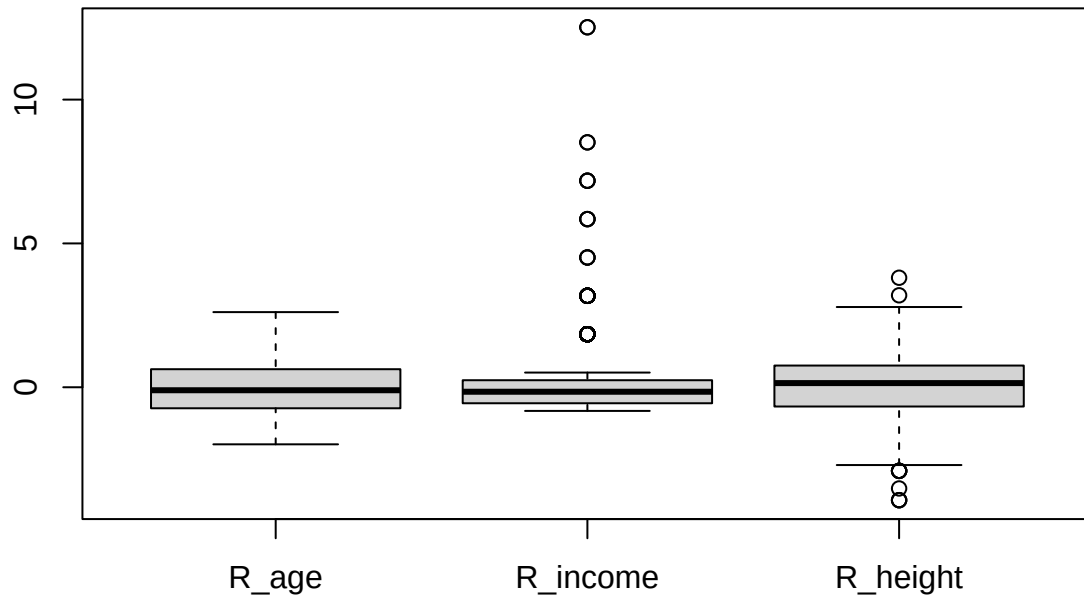
```
summarySE(ejercicio2,measurevar = "R_income")
```

```
##      .id      N R_income      sd      se      ci
## 1 <NA> 1534 617.0013 749.7612 19.14301 37.54926
```

```
summarySE(ejercicio2,measurevar = "R_height")
```

```
##      .id      N R_height      sd      se      ci
## 1 <NA> 1534 159.2907 4.914688 0.1254825 0.2461356
```

```
numuj<-ejercicio2[,c(2,4,6)]
boxplot(scale(numuj))
```



Lo que nos muestra esto es que las medias y medianas de las variables numéricas continuas son relativamente cercanas entre si en la mayoría de los casos.

Para realizar el boxplot lo que se hizo fue separar las columnas numéricas de las categóricas en otro objeto, así mismo se excluyeron aquellas mediciones referentes a las parejas de las mujeres entrevistadas. Lo que nos muestra la gráfica es que los grupos no son tan separados, pero hay muchas observaciones atípicas que exceden el rango de las variables ya sea con valores muy altos o muy bajos.

Análisis de conglomerados

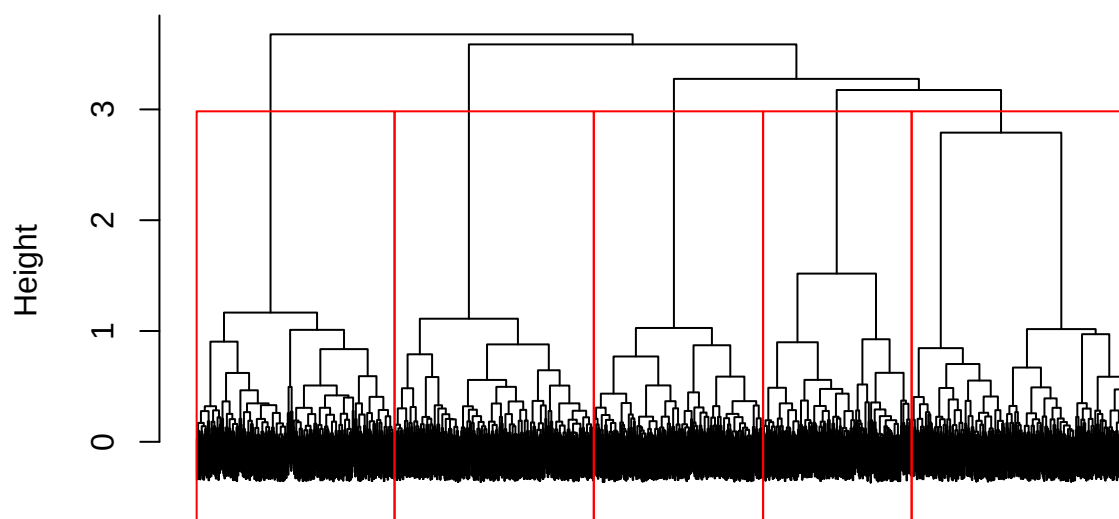
Para empezar con esto se limpió la base de datos para omitir los NAs, también se utilizó la distancia de Gower dado que en la base de datos se cuenta con variables de distintos tipos.

```
df_limpio<-ejercicio2 %>%
  na.omit() %>%
  mutate(across(c(R_health,R_happy,R_region,R_edu)))
dis_gow<-daisy(df_limpio,metric="gower")
```

Para realizar el árbol se utilizó el método Ward.D2 para evitar el encadenamiento y para compactar el árbol.

```
hc_arbol<-hclust(dis_gow,method="ward.D2")
plot(hc_arbol,main="tipos de mujeres",labels=F)
rect.hclust(hc_arbol, k=5, border="red")
```

tipos de mujeres



```
dis_gow
hclust (*, "ward.D2")
```

De acuerdo al resultado se observaron 5 grupos los cuales fueron remarcados con la función `rect.hclust`

```
grupitos<-cutree(hc_arbol, k=5)
ej4final<- cbind(df_limpio, Cluster=as.factor(grupitos))
```

Esto se realiza con el fin de etiquetar cada mujer dentro de 1 de los 5 grupos.

```
perfil_prom<-aggregate(cbind(R_age, R_income, R_height, as.numeric(R_health)) ~ Cluster,
                        data = ej4final,
                        FUN = mean, na.rm = TRUE)
conteos<-table(ej4final$Cluster)
perfil_prom
```

```
##   Cluster   R_age R_income R_height   V4
## 1      1 40.00722 439.4585 160.4188 3.534296
## 2      2 39.90535 674.4033 159.4198 3.699588
## 3      3 38.21607 367.2299 159.3684 3.808864
## 4      4 35.92593 852.1296 158.1821 3.716049
## 5      5 41.22699 773.0982 159.3037 3.809816
```

```
conteos
```

```
##
##   1   2   3   4   5
## 277 243 361 324 326
```


La función *aggregate* agrupa a cada mujer dentro de su cluster correspondiente y calcula su promedio de edad, ingresos, altura y salud. El objeto *perfil_prom* te da como resultado una tabla donde puedes ver los 5 clusters así como las variables de edad, ingresos, altura y salud con cada uno de sus promedios, mientras que el objeto *conteos* se trata de cuantas mujeres están dentro de estos grupos.

Ejercicio 3

Para este ejercicio se hará un MANOVA, lo que nos pide es determinar si a través de las 5 épocas hubo algún cambio en las medidas craneales para determinar si eso basta como evidencia de que existió el mestizaje. Se hizo un MANOVA porque realizar un ANOVA para cada variable aumentaría la probabilidad de cometer un error tipo 1, por lo que el manova nos permite realizar múltiples pruebas en una línea. Primero se hace un análisis exploratorio.

Resumen numérico y gráfico

```
tapply(calaveras,calaveras$epoch,summary)
```

```
## $c4000BC
##      epoch      mb      bh      bl      nh
## c4000BC:30  Min.   :119.0  Min.   :121.0  Min.   : 89.00  Min.   :44.00
## c3300BC: 0   1st Qu.:128.0  1st Qu.:131.2  1st Qu.: 95.00  1st Qu.:49.00
## c1850BC: 0   Median :131.0  Median :134.0  Median :100.00  Median :50.00
## c200BC : 0   Mean    :131.4  Mean    :133.6  Mean    : 99.17  Mean    :50.53
## cAD150 : 0   3rd Qu.:134.8  3rd Qu.:136.0  3rd Qu.:102.75  3rd Qu.:53.00
##              Max.    :141.0  Max.    :143.0  Max.    :114.00  Max.    :56.00
##
## $c3300BC
##      epoch      mb      bh      bl      nh
## c4000BC: 0   Min.   :123.0  Min.   :124.0  Min.   : 90.00  Min.   :45.00
## c3300BC:30  1st Qu.:130.0  1st Qu.:129.2  1st Qu.: 97.00  1st Qu.:48.00
## c1850BC: 0   Median :132.0  Median :133.0  Median : 98.50  Median :50.50
## c200BC : 0   Mean    :132.4  Mean    :132.7  Mean    : 99.07  Mean    :50.23
## cAD150 : 0   3rd Qu.:134.8  3rd Qu.:136.0  3rd Qu.:101.75  3rd Qu.:52.75
##              Max.    :148.0  Max.    :145.0  Max.    :107.00  Max.    :56.00
##
## $c1850BC
##      epoch      mb      bh      bl      nh
## c4000BC: 0   Min.   :126.0  Min.   :123.0  Min.   : 87.00  Min.   :45.00
## c3300BC: 0   1st Qu.:132.2  1st Qu.:131.0  1st Qu.: 92.25  1st Qu.:48.25
## c1850BC:30  Median :136.0  Median :133.5  Median : 96.00  Median :50.00
## c200BC : 0   Mean    :134.5  Mean    :133.8  Mean    : 96.03  Mean    :50.57
## cAD150 : 0   3rd Qu.:137.0  3rd Qu.:137.0  3rd Qu.: 99.75  3rd Qu.:52.75
##              Max.    :140.0  Max.    :145.0  Max.    :106.00  Max.    :60.00
##
## $c200BC
##      epoch      mb      bh      bl      nh
## c4000BC: 0   Min.   :129.0  Min.   :120.0  Min.   : 86.00  Min.   :46.00
## c3300BC: 0   1st Qu.:132.2  1st Qu.:130.0  1st Qu.: 91.25  1st Qu.:50.25
## c1850BC: 0   Median :135.0  Median :132.0  Median : 94.50  Median :52.00
## c200BC :30  Mean    :135.5  Mean    :132.3  Mean    : 94.53  Mean    :51.97
## cAD150 : 0   3rd Qu.:138.8  3rd Qu.:135.8  3rd Qu.: 97.75  3rd Qu.:53.75
```

```
##           Max.    :144.0   Max.    :142.0   Max.    :107.00   Max.    :60.00
##
## $cAD150
##      epoch      mb      bh      bl      nh
## c4000BC: 0   Min.    :126.0   Min.    :120.0   Min.    : 81.0   Min.    :44.00
## c3300BC: 0   1st Qu.:132.2   1st Qu.:126.0   1st Qu.: 91.0   1st Qu.:48.25
## c1850BC: 0   Median :137.0   Median :130.0   Median : 94.0   Median :52.00
## c200BC : 0   Mean    :136.2   Mean    :130.3   Mean    : 93.5   Mean    :51.37
## cAD150 :30   3rd Qu.:139.0   3rd Qu.:135.0   3rd Qu.: 97.0   3rd Qu.:54.00
##           Max.    :147.0   Max.    :138.0   Max.    :103.0   Max.    :58.00
```

```
#normalidad
```

```
tapply(calaveras$mb,calaveras$epoch,shapiro.test)
```

```
## $c4000BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.98136, p-value = 0.8603
##
##
## $c3300BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.92675, p-value = 0.04029
##
##
## $c1850BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.92262, p-value = 0.03139
##
##
## $c200BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.96262, p-value = 0.3608
##
##
## $cAD150
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.9792, p-value = 0.8039
```

```
tapply(calaveras$bh,calaveras$epoch,shapiro.test)
```

```
## $c4000BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.95664, p-value = 0.2536
##
##
```

```
## $c3300BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.97146, p-value = 0.5799
##
##
```

```
## $c1850BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.98322, p-value = 0.9031
##
##
```

```
## $c200BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.98307, p-value = 0.9
##
##
```

```
## $cAD150
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.93745, p-value = 0.07767
```

```
tapply(calaveras$bl,calaveras$epoch,shapiro.test)
```

```
## $c4000BC
##
## Shapiro-Wilk normality test
##
## data:  X[[i]]
## W = 0.97314, p-value = 0.6282
##
##
## $c3300BC
```

```
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.97728, p-value = 0.7495
##
##
## $c1850BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.98239, p-value = 0.8849
##
##
## $c200BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.96957, p-value = 0.5276
##
##
## $cAD150
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.97811, p-value = 0.7734
```

```
tapply(calaveras$nh,calaveras$epoch,shapiro.test)
```

```
## $c4000BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.97481, p-value = 0.6772
##
##
## $c3300BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.96335, p-value = 0.3762
##
##
## $c1850BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.9639, p-value = 0.3881
```

```
##
##
## $c200BC
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.95638, p-value = 0.2496
##
##
## $cAD150
##
## Shapiro-Wilk normality test
##
## data: X[[i]]
## W = 0.97355, p-value = 0.6401
```

```
#summarySE
library(Rmisc)
summarySE(calaveras,measurevar="mb",groupvars="epoch")
```

```
##      epoch  N      mb      sd      se      ci
## 1 c4000BC 30 131.3667 5.129249 0.9364685 1.915293
## 2 c3300BC 30 132.3667 4.810071 0.8781948 1.796110
## 3 c1850BC 30 134.4667 3.481313 0.6355979 1.299944
## 4 c200BC 30 135.5000 3.919448 0.7155899 1.463546
## 5 cAD150 30 136.1667 5.350368 0.9768391 1.997860
```

```
summarySE(calaveras,measurevar="bh",groupvars="epoch")
```

```
##      epoch  N      bh      sd      se      ci
## 1 c4000BC 30 133.6000 4.469051 0.8159333 1.668771
## 2 c3300BC 30 132.7000 4.647209 0.8484604 1.735296
## 3 c1850BC 30 133.8000 4.978575 0.9089592 1.859030
## 4 c200BC 30 132.3000 5.133729 0.9372864 1.916966
## 5 cAD150 30 130.3333 4.971181 0.9076094 1.856270
```

```
summarySE(calaveras,measurevar="bl",groupvars="epoch")
```

```
##      epoch  N      bl      sd      se      ci
## 1 c4000BC 30 99.16667 5.884423 1.0743438 2.197280
## 2 c3300BC 30 99.06667 4.346488 0.7935564 1.623005
## 3 c1850BC 30 96.03333 4.552251 0.8311235 1.699838
## 4 c200BC 30 94.53333 4.591847 0.8383527 1.714624
## 5 cAD150 30 93.50000 5.056576 0.9232003 1.888157
```

```
summarySE(calaveras,measurevar="nh",groupvars="epoch")
```

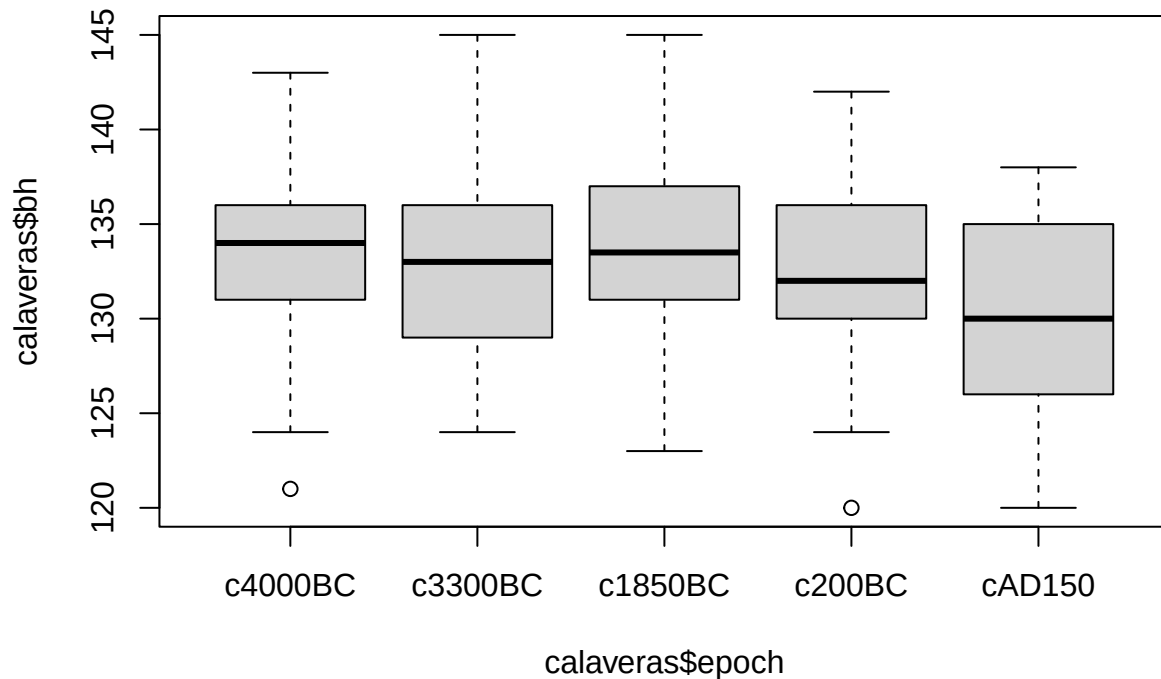
```
##      epoch  N      nh      sd      se      ci
## 1 c4000BC 30 50.53333 2.763473 0.5045388 1.031898
## 2 c3300BC 30 50.23333 2.955805 0.5396536 1.103716
```

```
## 3 c1850BC 30 50.56667 3.549486 0.6480445 1.325400
## 4 c200BC 30 51.96667 2.822121 0.5152465 1.053797
## 5 cAD150 30 51.36667 3.718392 0.6788823 1.388470
```

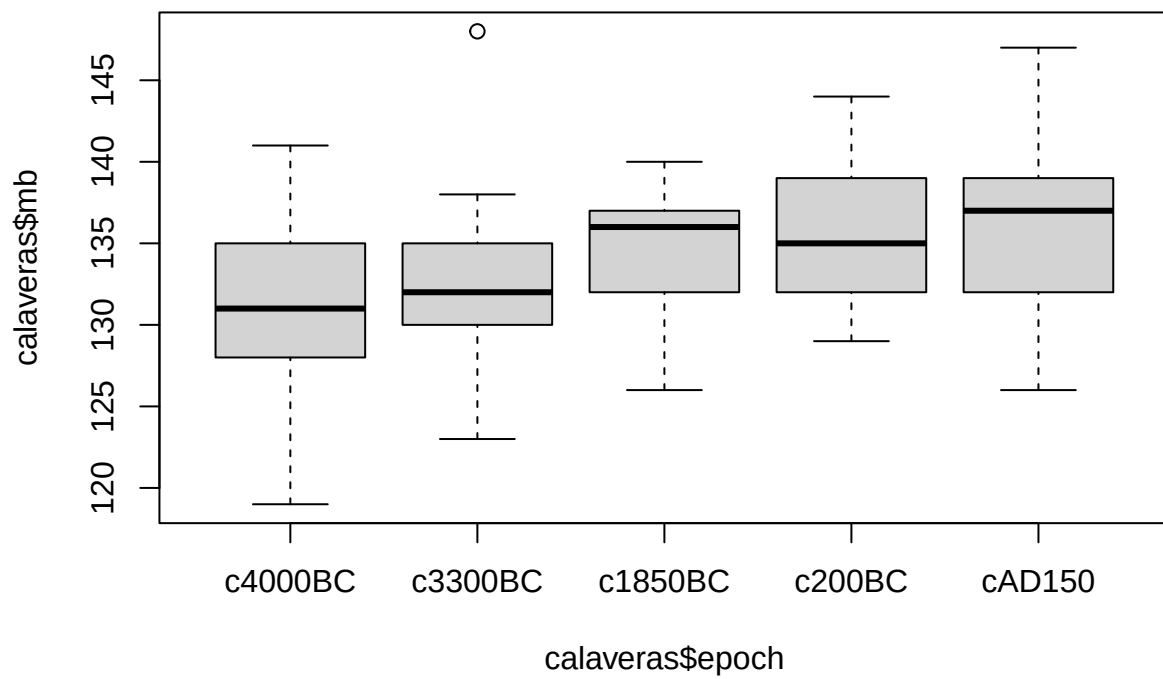
Aquí se puede observar que las medianas y medias son cercanas entre sí, además las pruebas de normalidad en todas no se rechaza la hipótesis nula, con dos excepciones y de igual manera las desviaciones estándar son muy cercanas entre si con todas las variables.

Para los boxplots, en todos los casos no se puede observar a simple vista alguna diferencia entre las variables, todos rangos se sobrelapan entre si al igual que las medianas.

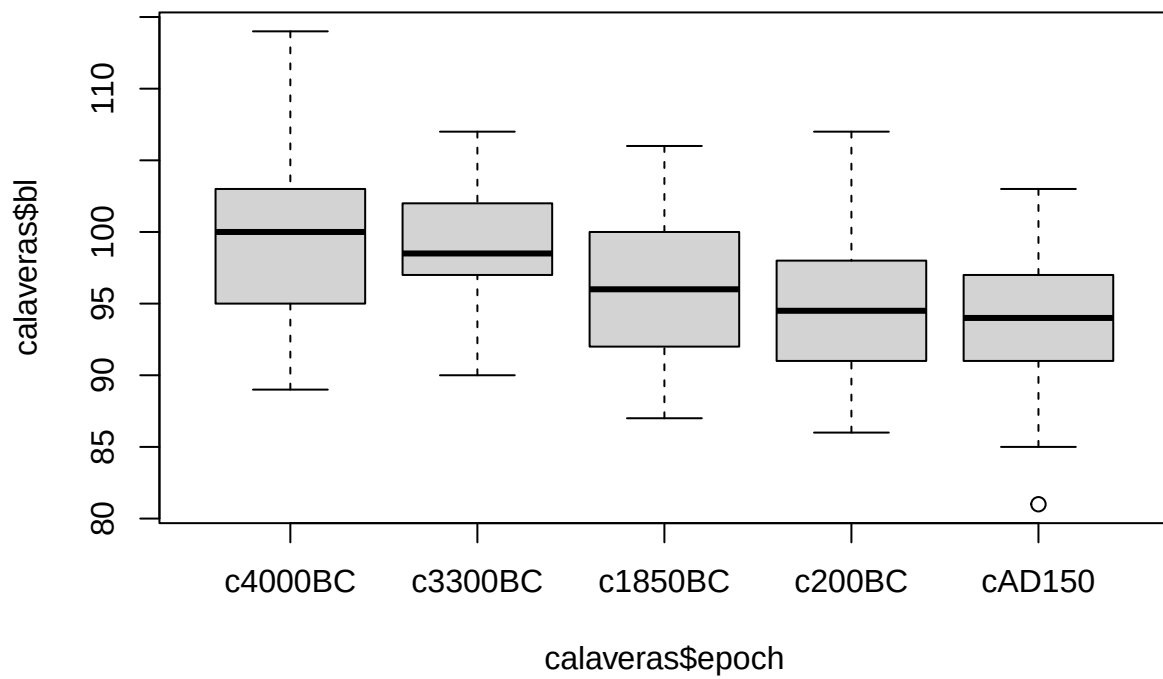
```
boxplot(calaveras$bh~calaveras$epoch)
```



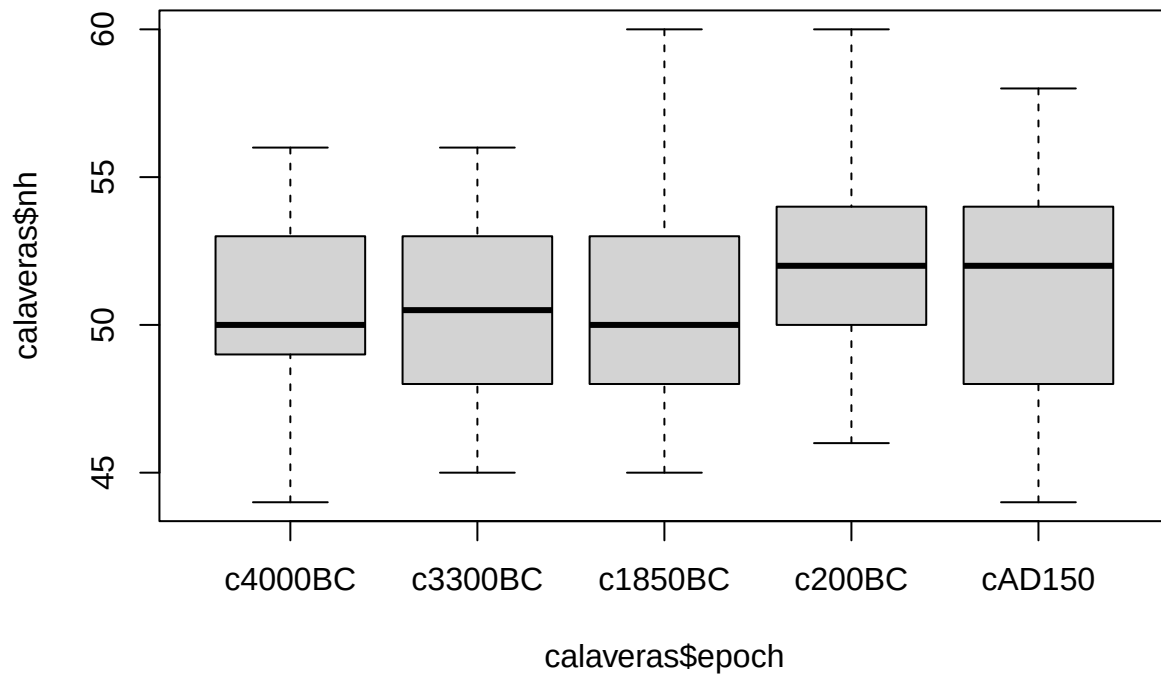
```
boxplot(calaveras$mb~calaveras$epoch)
```



```
boxplot(calaveras$mb~calaveras$epoch)
```



```
boxplot(calaveras$nh~calaveras$epoch)
```

Primero se hace un ANOVA para verificar los supuestos

```
model1<-aov(calaveras$mb~calaveras$epoch)
summary(model1)
```

```
##               Df Sum Sq Mean Sq F value    Pr(>F)
## calaveras$epoch   4  502.8   125.71     5.955 0.000183 ***
## Residuals       145 3061.1    21.11
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
shapiro.test(model1$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  model1$residuals
## W = 0.99328, p-value = 0.7116
```

```
bartlett.test(calaveras$mb~calaveras$epoch)#si todo bien
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  calaveras$mb by calaveras$epoch
## Bartlett's K-squared = 7.3382, df = 4, p-value = 0.1191
```

```
TukeyHSD(model)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = calaveras$mb ~ calaveras$epoch)
##
## $'calaveras$epoch'
##          diff          lwr          upr          p adj
## c3300BC-c4000BC 1.0000000 -2.2771344 4.277134 0.9166997
## c1850BC-c4000BC 3.1000000 -0.1771344 6.377134 0.0732388
## c200BC-c4000BC 4.1333333 0.8561989 7.410468 0.0058079
## cAD150-c4000BC 4.8000000 1.5228656 8.077134 0.0007938
## c1850BC-c3300BC 2.1000000 -1.1771344 5.377134 0.3949737
## c200BC-c3300BC 3.1333333 -0.1438011 6.410468 0.0682844
## cAD150-c3300BC 3.8000000 0.5228656 7.077134 0.0142258
## c200BC-c1850BC 1.0333333 -2.2438011 4.310468 0.9070646
## cAD150-c1850BC 1.7000000 -1.5771344 4.977134 0.6074686
## cAD150-c200BC 0.6666667 -2.6104677 3.943801 0.9802743
```

De acuerdo a los resultados estos dos grupos si tienen diferencias entre sí dado el valor de p, los residuales se distribuyen de manera normal, hay homogeneidad de varianza y se realizó la prueba de Tukey la cual muestra que la primera época es diferente a la cuarta al igual que la última época, y la última época es diferente a la segunda.

Para hacer el MANOVA primero se asignan todas las variables de respuestas a un vector y se empieza a hacer el modelo con la única variable explicativa que en este caso son las épocas.

```
respuestas<-cbind(calaveras$mb,calaveras$bh,calaveras$bl,calaveras$nh)
mano_va<-manova(respuestas~epoch,data=calaveras)
summary(mano_va)
```

```
##          Df Pillai approx F num Df den Df      Pr(>F)
## epoch      4 0.35331    3.512     16    580 4.675e-06 ***
## Residuals 145
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Posteriormente se realiza un resumen de anova para observar cada uno de los anovas y se comprueba el supuesto de normalidad.

```
shapiro.test(mano_va$residuals)
```

```
##
## Shapiro-Wilk normality test
##
## data:  mano_va$residuals
## W = 0.9965, p-value = 0.2142
```

```
summary.aov(mano_va)
```

```
## Response 1 :
##           Df Sum Sq Mean Sq F value    Pr(>F)
## epoch           4  502.83  125.707   5.9546 0.0001826 ***
## Residuals    145 3061.07   21.111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response 2 :
##           Df Sum Sq Mean Sq F value    Pr(>F)
## epoch           4  229.9   57.477   2.4474 0.04897 *
## Residuals    145 3405.3   23.485
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response 3 :
##           Df Sum Sq Mean Sq F value    Pr(>F)
## epoch           4  803.3  200.823   8.3057 4.636e-06 ***
## Residuals    145 3506.0   24.179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response 4 :
##           Df Sum Sq Mean Sq F value    Pr(>F)
## epoch           4   61.2   15.300   1.507 0.2032
## Residuals    145 1472.1   10.153
```

Finalmente se hace un análisis de discriminantes.

```
mod_lda<-lda(epoch~mb+bh+bl+nh,data=calaveras)
mod_lda
```

```
## Call:
## lda(epoch ~ mb + bh + bl + nh, data = calaveras)
##
## Prior probabilities of groups:
## c4000BC c3300BC c1850BC c200BC cAD150
##      0.2      0.2      0.2      0.2      0.2
##
## Group means:
##           mb           bh           bl           nh
## c4000BC 131.3667 133.6000 99.16667 50.53333
## c3300BC 132.3667 132.7000 99.06667 50.23333
## c1850BC 134.4667 133.8000 96.03333 50.56667
## c200BC  135.5000 132.3000 94.53333 51.96667
## cAD150  136.1667 130.3333 93.50000 51.36667
##
## Coefficients of linear discriminants:
##           LD1           LD2           LD3           LD4
## mb  0.12667629  0.03873784 -0.09276835 -0.1488398644
## bh -0.03703209  0.21009773  0.02456846  0.0004200843
## bl -0.14512512 -0.06811443 -0.01474860 -0.1325007670
## nh  0.08285128 -0.07729281  0.29458931 -0.0668588797
##
```

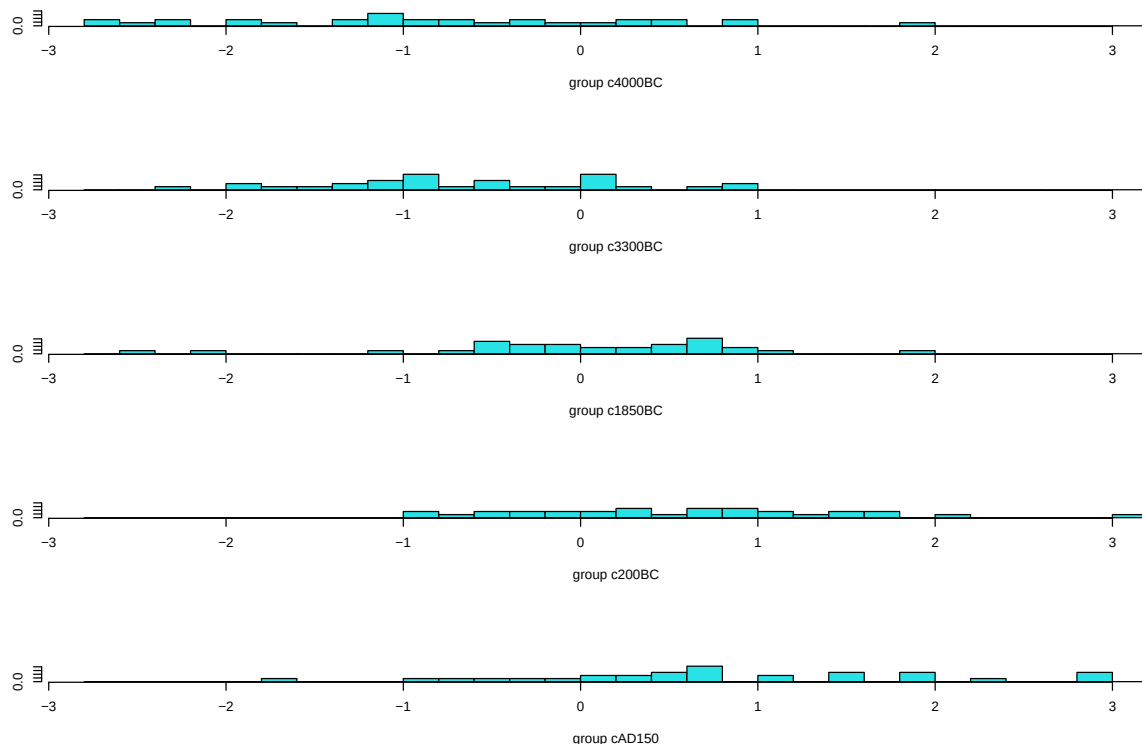
```
## Proportion of trace:
##   LD1   LD2   LD3   LD4
## 0.8823 0.0809 0.0326 0.0042
```

```
#plot(mod_lda, dimen=1)
```

Se hace con todas las variables de respuesta y se asigna a un objeto llamado *mod_lda*, este nos da como resultado la proporción de cada discriminante, el cual podemos ver que el primero tiene un 88% de variación por lo que solo ese podría ser necesario para graficar.

En la gráfica se puede observar que efectivamente existen cambios en las medidas craneales en las épocas teniendo evidencia de que el mestizaje tuvo lugar en esa población. La gráfica no pudo cargar pero se adjunta una imagen del visualizador para ejemplificar el resultado.

```
knitr::include_graphics("C:/Users/100032608/Documents/VERANO-MULTI/plot_mod_lda.pdf")
```



Ejercicio 4

Para este ejercicio se hará un escalamiento multidimensional no paramétrico, porqueal tener como entrada una matriz de disimilitud, con este método se podrán representar de mejor manera las relaciones descritas ahí. Para empezar se transforman los datos porque el problema nos indica que tenemos porcentajes, se usará la transformación de Hellinger. Después se hará una matriz de disimilitud con el método euclidiano.

En los argumentos se usa $k=2$ para reducir las dimensiones del modelo evitando su complejidad, *trymax=100* se refiere al número de veces con que va a mapear los puntos para que al final se quede con aquel modelo que tenga el *stress* más bajo.

```

watervoles1<-decostand(watervoles,method="hellinger")
matridist<-vegdist(watervoles1,method="euclidean")
ahorasi<-metaMDS(matridist,k=2,trymax=100)

```

```

## Run 0 stress 0.0412489
## Run 1 stress 0.0412489
## ... New best solution
## ... Procrustes: rmse 6.551931e-05  max resid 0.0001814417
## ... Similar to previous best
## Run 2 stress 0.04124893
## ... Procrustes: rmse 4.898228e-05  max resid 0.0001342606
## ... Similar to previous best
## Run 3 stress 0.04124889
## ... New best solution
## ... Procrustes: rmse 3.460363e-05  max resid 9.214955e-05
## ... Similar to previous best
## Run 4 stress 0.04124889
## ... New best solution
## ... Procrustes: rmse 2.355146e-05  max resid 5.924163e-05
## ... Similar to previous best
## Run 5 stress 0.0412489
## ... Procrustes: rmse 5.520705e-05  max resid 0.0001517762
## ... Similar to previous best
## Run 6 stress 0.04124889
## ... New best solution
## ... Procrustes: rmse 8.091327e-06  max resid 2.08953e-05
## ... Similar to previous best
## Run 7 stress 0.04124894
## ... Procrustes: rmse 0.0001053081  max resid 0.000245083
## ... Similar to previous best
## Run 8 stress 0.04124898
## ... Procrustes: rmse 0.0001428528  max resid 0.0003758396
## ... Similar to previous best
## Run 9 stress 0.04124889
## ... Procrustes: rmse 5.027286e-06  max resid 1.120011e-05
## ... Similar to previous best
## Run 10 stress 0.08118006
## Run 11 stress 0.0412489
## ... Procrustes: rmse 3.624362e-05  max resid 9.992699e-05
## ... Similar to previous best
## Run 12 stress 0.04124893
## ... Procrustes: rmse 7.062089e-05  max resid 0.0001430097
## ... Similar to previous best
## Run 13 stress 0.0822802
## Run 14 stress 0.04124891
## ... Procrustes: rmse 4.117449e-05  max resid 0.0001120301
## ... Similar to previous best
## Run 15 stress 0.04124894
## ... Procrustes: rmse 8.253413e-05  max resid 0.000222244
## ... Similar to previous best
## Run 16 stress 0.04124897
## ... Procrustes: rmse 0.0001169359  max resid 0.0003457063
## ... Similar to previous best

```

```
## Run 17 stress 0.0412489
## ... Procrustes: rmse 4.671752e-05  max resid 0.0001318351
## ... Similar to previous best
## Run 18 stress 0.04124889
## ... Procrustes: rmse 1.468812e-05  max resid 4.02717e-05
## ... Similar to previous best
## Run 19 stress 0.04124892
## ... Procrustes: rmse 6.261549e-05  max resid 0.0001410765
## ... Similar to previous best
## Run 20 stress 0.04124889
## ... Procrustes: rmse 2.808196e-05  max resid 6.867679e-05
## ... Similar to previous best
## *** Best solution repeated 13 times
```

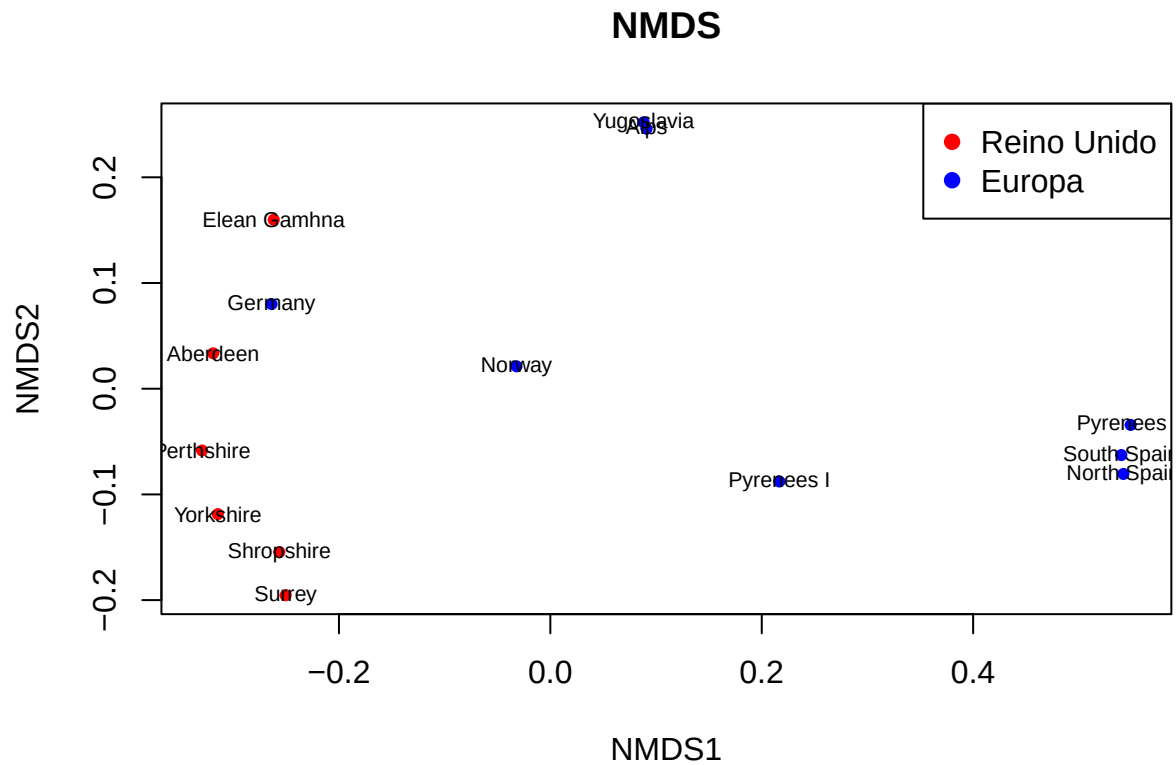
Para el objeto *grups* se crean etiquetas para que los primeros 6 datos estén en la categoría de Reino Unido y los siguientes 8 a Europa correspondiendo a la descripción de los datos.

Las siguientes líneas son para poner los puntos y texto en la gráfica para poder determinar si hay una separación en los datos.

```
grups<-c(rep("Reino Unido",6),rep("Europa",8))
plot(ahorasi,type="n",main="NMDS")
```

```
## species scores not available
```

```
points(ahorasi,display="sites",col=ifelse(grups=="Reino Unido","red","blue"),pch=19)
text(ahorasi,labels=rownames(watervoles))
legend("topright",legend=c("Reino Unido","Europa"),col=c("red","blue"),pch=19)
```



En el gráfico se puede observar que los datos si están separados en su mayoría, hay valores azules que corresponden a Europa que están situados muy a la derecha mientras que los de Reino Unido (rojos) están a la izquierda, sin embargo, hay un punto donde el punto correspondiente a Alemania se mezcla con los demás de Reino Unido.

Los puntos de Europa están distribuidos ampliamente lo que permite que se lleguen a mezclar.